Table 1: Win rate of Baguan-TS (ensemble) against baselines on `fev-bench-cov`, broken down by dataset category. Energy + nature datasets include ENTSOE, Solar, EPF, and UCI Air Quality; retail datasets include Rossmann, Favorita, Walmart, M5, Rohlik, and Hermes.

All 30 Datasets		**15 Energy & Nature Datasets**		**15 Retail Datasets**	
Baseline	**Win Rate**	**Baseline**	**Win Rate**	**Baseline**	**Win Rate**
Chronos-Bolt	0.80	Chronos-Bolt	0.87	Chronos-Bolt	0.73
Moirai-2.0	0.83	Moirai-2.0	0.93	Moirai-2.0	0.73
Sundial-Base	0.93	Sundial-Base	0.87	Sundial-Base	1.00
TimesFM-2.5	0.50	TimesFM-2.5	0.67	TimesFM-2.5	0.33
TabPFN-TS	0.67	TabPFN-TS	0.73	TabPFN-TS	0.60
TiRex	0.70	TiRex	0.87	TiRex	0.53
TimesFM-2.5 (excl. Proenfo)*	0.56	TimesFM-2.5 (excl. Proenfo)*	0.83	—	—

* TimesFM-2.5 may have been pre-trained on data that overlaps with the Proenfo datasets, potentially leading to data leakage. We therefore additionally report win rates on the remaining 27 datasets (12 energy and nature datasets) with the 3 Proenfo datasets excluded.

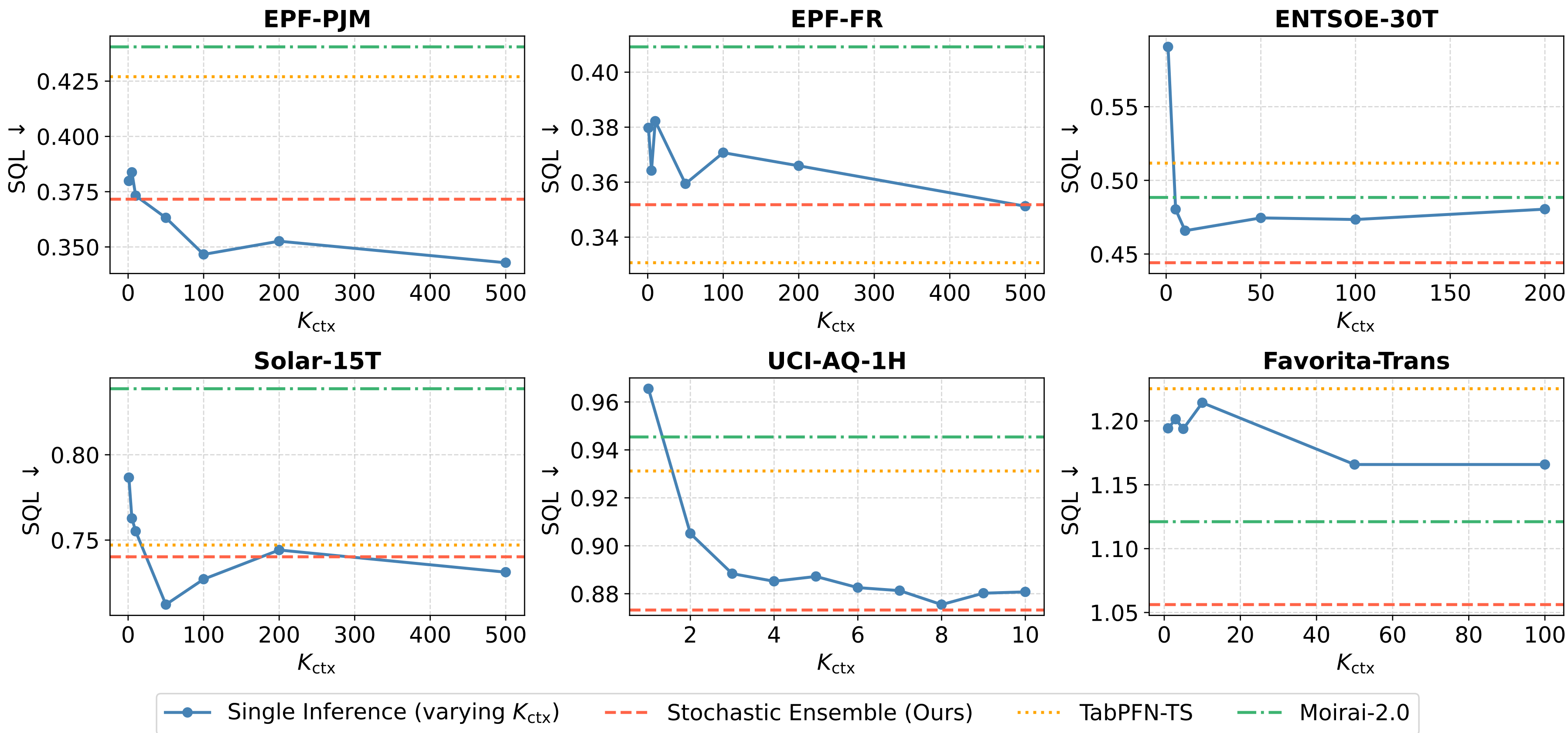

Figure 1: Parameter sensitivity analysis of the number of retrieved context slices (K_{ctx}) across six datasets. The blue curve shows the SQL of Baguan-TS under single inference as K_{ctx} varies, while the horizontal dashed/dotted lines indicate the performance of Baguan-TS stochastic ensemble (ours), TabPFN-TS, and Moirai-2.0. Lower SQL is better. When K_{ctx} is too small, the model suffers from insufficient contextual information, leading to unstable performance. As K_{ctx} increases, richer retrieved context generally improves forecasting accuracy; however, excessively large K_{ctx} may introduce redundant or less relevant context, resulting in diminishing or fluctuating gains.

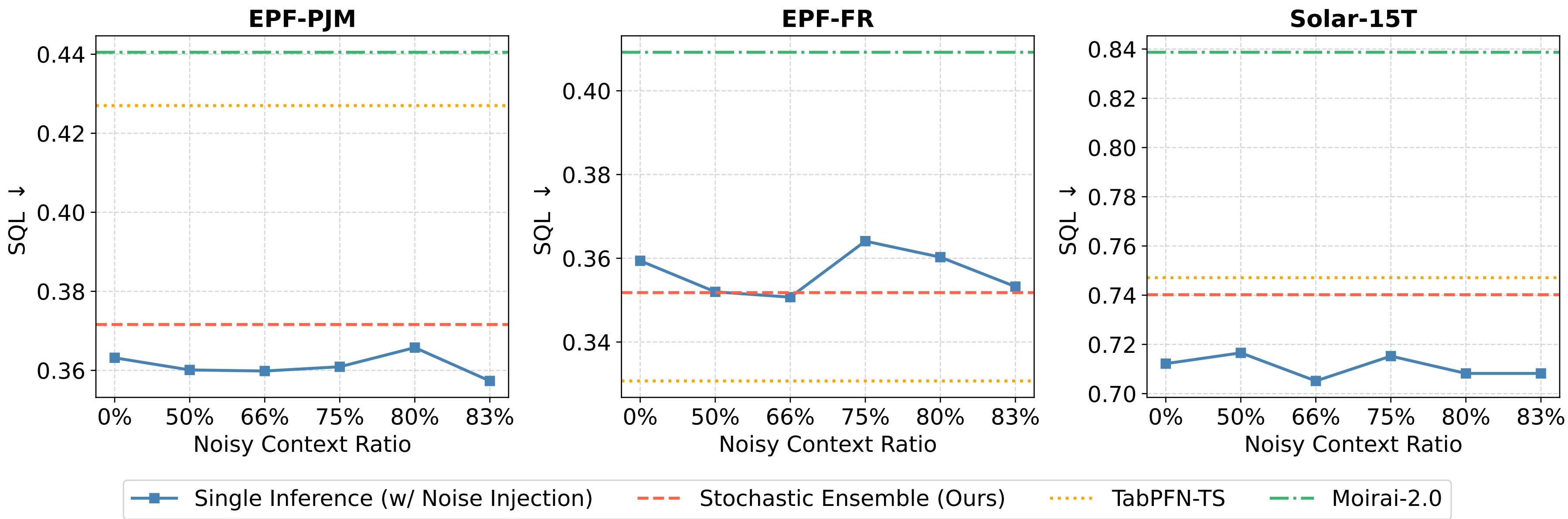

Figure 2: Robustness of Baguan-TS to noisy retrieval context at inference time. We inject different proportions of random context slices into the retrieved context and report the SQL of Baguan-TS under single inference on three datasets. The horizontal dashed/dotted lines indicate the performance of Baguan-TS stochastic ensemble (ours), TabPFN-TS, and Moirai-2.0. Lower SQL is better. Baguan-TS is robust to imperfect retrieval: a small amount of injected noise can even slightly improve performance, whereas larger noise ratios lead to only moderate degradation, demonstrating stable and controllable behavior under noisy context.

Table 2: Efficiency–accuracy comparison on 9 photovoltaic datasets. Lower is better for all forecasting metrics. Results are averaged over 5 runs and reported as mean ± std.

Model	Latency (s/sample)	Peak GPU Mem (MB)	SQL ↓	MASE ↓	WAPE ↓	WQL ↓
Baguan-TS (full ensemble)	3.672	2715.1	0.3757 ± 0.0057	0.4533 ± 0.0092	0.1361 ± 0.0028	0.1126 ± 0.0017
Baguan-TS (single-forward)	0.427	1440.1	0.3810	0.4597	0.1380	0.1142
Chronos-Bolt	0.043	226.2	0.6284	0.7795	0.2344	0.1889
MOIRAI2	0.049	4.9	0.6175	0.7222	0.2175	0.1859
TimesFM-2.5	0.310	1368.6	0.6434	0.7572	0.2266	0.1925
ToTO	0.235	2914.2	0.7484 ± 0.0076	0.8993 ± 0.0095	0.2696 ± 0.0028	0.2245 ± 0.0023
Sundial	0.060	573.1	0.7281 ± 0.0104	0.8216 ± 0.0097	0.2472 ± 0.0029	0.2191 ± 0.0031
TabPFN-TS	4.481	430.6	0.3766	0.4728	0.1421	0.1132
TiRex	2.532	–	0.6042	0.7490	0.2253	0.1813

Table 3: Per-dataset SQL comparison on 6 datasets from `fev-bench-cov`. Baguan-TS is evaluated under a single fixed configuration, with and without covariate information. Lower SQL is better (↓).

Method	EPF-PJM	EPF-FR	ENTSOE-30T	Solar-15T	UCI-AQ-1H	Fav.-Trans	Average
Baguan-TS (w/ covariates)	**0.3632**	0.3594	**0.4745**	**0.7122**	0.8852	**1.1659**	**0.6601**
Baguan-TS (w/o covariates)	0.3944	0.4586	0.4899	0.7634	**0.8712**	1.1860	0.6939
TabPFN-TS	0.4270	**0.3307**	0.5117	0.7471	0.9312	1.2252	0.6955
TimesFM-2.5	0.4263	0.4092	0.5658	0.9063	0.8769	0.8736	0.6764
Chronos-Bolt	0.4217	0.5638	0.5294	0.8094	0.8990	0.9750	0.6997
Moirai-2.0	0.4405	0.5035	0.4884	0.8387	0.9454	1.1211	0.7229
Sundial-Base	0.4611	0.4679	0.7216	0.9626	1.0009	1.1982	0.8021
TiRex	0.4042	0.4014	0.5230	0.8457	0.8650	1.0314	0.6785

Table 4: Performance comparison on 9 photovoltaic datasets under the known-future-covariate setting, including pretrained zero-shot foundation models and covariate-aware supervised baselines trained from scratch. Lower is better for all metrics.

Model	Type	SQL ↓	MASE ↓	WAPE ↓	WQL ↓
BaguanTS	Zero-shot FM	0.376	0.453	0.136	0.113
TabPFN-TS	Zero-shot FM	0.377	0.473	0.142	0.114
Chronos-Bolt	Zero-shot FM	0.628	0.779	0.234	0.189
Moirai-2	Zero-shot FM	0.618	0.722	0.227	0.186
TFT	Task-specific	0.715	0.757	0.226	0.213
DeepAR	Task-specific	1.144	1.144	0.340	0.340
N-HiTS	Task-specific	2.281	2.390	0.711	0.678

Table 5: Win rate of Baguan-TS against baselines on `fev-bench-cov`, broken down by dataset category, under two ensemble strategies: search-based ensemble (per-dataset lookback window search with validation-based selection, as reported in the original submission) and fixed-config ensemble (grid over inference mode, RevIN, and context window length without selection).

All 30 Datasets			**15 Energy & Nature Datasets**			**15 Retail Datasets**		
Baseline	**Search-Based**	**Fixed-Config**	**Baseline**	**Search-Based**	**Fixed-Config**	**Baseline**	**Search-Based**	**Fixed-Config**
Chronos-Bolt	0.80	0.83	Chronos-Bolt	0.87	1.00	Chronos-Bolt	0.73	0.67
Moirai-2.0	0.83	0.77	Moirai-2.0	0.93	0.93	Moirai-2.0	0.73	0.60
Sundial-Base	0.93	0.90	Sundial-Base	0.87	0.87	Sundial-Base	1.00	0.93
TimesFM-2.5	0.50	0.47	TimesFM-2.5	0.67	0.67	TimesFM-2.5	0.33	0.27
TabPFN-TS	0.67	0.53	TabPFN-TS	0.73	0.60	TabPFN-TS	0.60	0.47
TiRex	0.70	0.60	TiRex	0.87	0.87	TiRex	0.53	0.33
TimesFM-2.5 (excl. Proenfo)	0.56	0.52	TimesFM-2.5 (excl. Proenfo)	0.83	0.83	—	—	—

Table 6: Fixed-configuration SQL and standard deviation for Baguan-TS on `fev-bench-cov`. SQL values are obtained from a fixed-configuration ensemble over all combinations of inference mode (2D, 3D), RevIN (enabled/disabled for 3D), and context window length (5×, 14× the dataset's seasonality), without any validation-based selection. Standard deviations are computed over 6 runs with different random seeds under a single fixed configuration (3D inference mode, RevIN enabled, context window 5× the dataset's seasonality), as inference stochasticity only manifests in the random masking of the 3D mode. The results demonstrate **strong stability** across datasets, with standard deviations consistently **below 0.01** for the majority of benchmarks, confirming that the stochasticity introduced by random masking and covariate shuffling during inference has minimal impact on predictive performance.

Dataset	SQL ↓	Std	Dataset	SQL ↓	Std
`proenfo_gfc12`	0.6886	0.0024	`rohlik_sales_1D`	0.9350	0.0006
`proenfo_gfc14`	0.5374	0.0040	`rohlik_orders_1D`	1.2385	0.0104
`proenfo_gfc17`	0.6388	0.0017	`rohlik_sales_1W`	1.2251	0.0024
`entsoe_15T`	0.4541	0.0037	`rohlik_orders_1W`	1.7394	0.0177
`entsoe_30T`	0.4663	0.0021	`favorita_stores_1W`	2.1016	0.0057
`entsoe_1H`	0.4252	0.0035	`favorita_stores_1M`	2.0058	0.0022
`solar_with_weather_15T`	0.7252	0.0072	`favorita_transactions_1D`	1.1931	0.0421
`solar_with_weather_1H`	0.7071	0.0056	`favorita_stores_1D`	0.9726	0.0003
`uci_air_quality_1H`	0.8945	0.0029	`rossmann_1W`	0.2731	0.0004
`uci_air_quality_1D`	1.0818	0.0084	`rossmann_1D`	0.2931	0.0001
`epf_be`	0.5344	0.0086	`hermes`	0.6395	0.0001
`epf_de`	0.4807	0.0045	`walmart`	0.7463	0.0008
`epf_fr`	0.3555	0.0032	`m5_1W`	0.9099	0.0006
`epf_np`	0.7977	0.0053	`m5_1M`	1.0058	0.0002
`epf_pjm`	0.3395	0.0066	`m5_1D`	0.7281	0.0001

Table 7: The 22 real-world datasets from the GIFT-Eval pretraining corpus used for training Baguan-TS.

Dataset	Source	Domain
`Beijing Subway`	LibCity	Transport
`HZMetro`	LibCity	Transport
`Los-Loop`	LibCity	Transport
`PEMS03`	LibCity	Transport
`PEMS04`	LibCity	Transport
`PEMS07`	LibCity	Transport
`PEMS08`	LibCity	Transport
`PEMS Bay`	LibCity	Transport
`Q-Traffic`	LibCity	Transport
`SHMetro`	LibCity	Transport
`Alibaba Cluster Trace 2018`	CloudOpsTSF	CloudOps
`Azure VM Traces 2017`	CloudOpsTSF	CloudOps
`Borg Cluster Data 2011`	CloudOpsTSF	CloudOps
`Australian Elec. Demand`	Monash	Energy
`Bitcoin`	Monash	Econ/Fin
`BDG-2 Bear`	BuildingsBench	Energy
`BDG-2 Fox`	BuildingsBench	Energy
`BDG-2 Panther`	BuildingsBench	Energy
`BDG-2 Rat`	BuildingsBench	Energy
`Borealis`	BuildingsBench	Energy
`Buildings900K`	BuildingsBench	Energy
`Beijing Air Quality`	LOTSA_Others	Nature